Predicting Market Volatility and Short-Term Trading Signals from Reddit's WallStreetBets

UE24CS352A Machine Learning - Mini-Project write-up | Team: Prithvi Aithal (PES1UG24CS909), Sagar A Walikar (PES1UG24CS396) | Repository: github.com/Prithvi-Aithal/wsb-market-prediction (private)

1. Problem statement

Retail-investor communities such as r/WallStreetBets (WSB) moved prices during the January 2021 GameStop short squeeze. We ask whether the **posts themselves** contain a usable signal about (a) the **direction** of the next price move (will open(next day) exceed open(this day)?), (b) **volatility** (will the move be unusually large?), and (c) whether such predictions can drive a simple short-term trading strategy. We re-implement the CS229 report's approach, extend it with hyper-parameter tuning, stricter baselines, a leakage-free evaluation and a back-test, and add a **new model** (a drift-aware shrinkage ensemble) to improve on the report's two models.

2. Dataset

Posts: Kaggle *gpreda/reddit-wallstreetsbets-posts* - 53,187 posts (Sep 2020 - Aug 2021) with title, score, id, url, comment count, creation time, body, timestamp. **Prices:** daily OHLC from Yahoo Finance (*yfinance*). Posts mentioning each stock: GME 14,619, AMC 5,310, TSLA 1,232, FB 251 (ticker or company name). Each post becomes one example, as in the report.

3. Approach

Pre-processing: drop NaN titles, lowercase, remove URLs, handles, special and single characters, collapse spaces. **Features (7)**: post_score, total_comms_num, len_of_post, positive/neutral/negative sentiment (VADER), bias term; count features are log1p-transformed and standardised on the training set. **Labels**: *direction* = open(k+1) > open(k); *volatility* = |open(k+1)/open(k)-1| above the training-set median (the report's 'significant shift'). Two timing conventions: *paper* (k = post's trading day) and *tradable* (k = first open after the post, so no look-ahead). **Models**: (1) logistic regression solved with **Newton's method implemented from scratch** (small L2 term keeps the Hessian invertible); (2) 3x256 ReLU network, sigmoid output, Adam (lr 1e-3), 30 epochs, batch 16 (64 for GME/TSLA), best epoch by validation accuracy; (3) uniform-random and majority-class baselines; (4) **new: shrinkage ensemble** - soft vote of Newton logistic regression, random forest, gradient boosting and Gaussian naive Bayes, blended with the training base rate: $p = \lambda\, p_{vote} + (1-\lambda)$ prior. Its count features are *relative* (percentile rank of score/comments/length among the stock's last 200 posts, only past posts used), because raw engagement drifts about 10x as the subreddit grew after the squeeze. λ (0 = majority rule, 1 = plain ensemble) and the base learners' hyper-parameters (random search, 8 configs: L2, forest leaf size, boosting trees/depth/learning rate) are chosen by **forward-chaining cross-validation** log-loss on train+val; the chosen model is refit on train+val. **Tuning**: logistic-regression L2 in {1e-6..10} by validation log-loss; network random search (6 configs: depth/width, learning rate, weight decay, batch) by validation accuracy, re-trained with 5 seeds. **Evaluation**: chronological 80/10/10 train/val/test split so the test set is strictly in the future.

4. Implementation overview

Python; *src/wsb_pipeline.py* (data, features, labels, splits), *src/models.py* (Newton logistic regression, network), *src/run_experiments.py* (experiments, tuning, baselines, back-test, figures), *src/leakage_check.py* (shuffled vs chronological split), *src/demo.py* (live scoring of a post, and a *--quick* mode that runs the whole pipeline live on one stock), *tests/* (unit tests). Back-test: for every trading day, majority vote of the day's per-post predictions -> long (open to next open) or flat, compared with going long on every day that had posts.

5. Results (held-out chronological test set, accuracy)

Direction task, 'tradable' labels

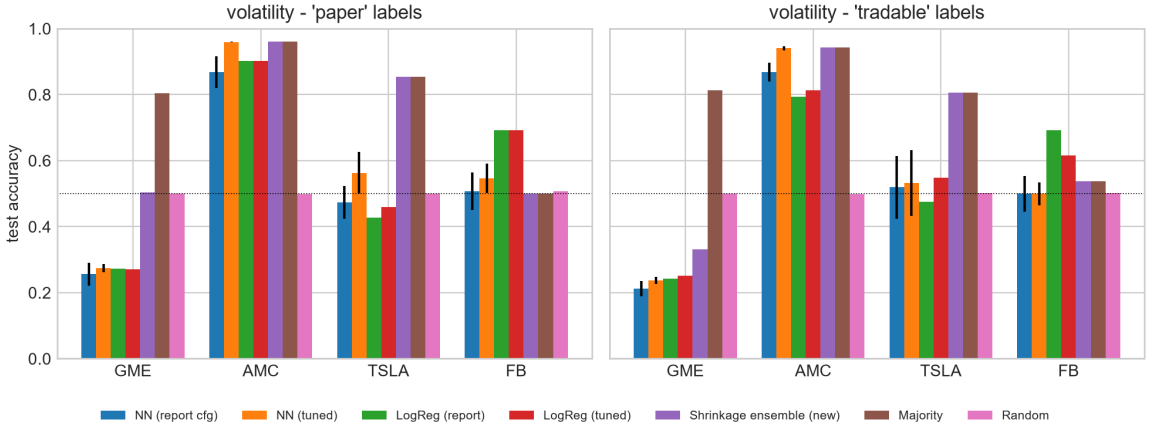
Stock	n test	NN (report)	NN (tuned)	LogReg (report)	LogReg (tuned)	Ensemble (new)	Majority	Random
GME	1462	0.569	0.569	0.567	0.567	0.568	0.567	0.500
AMC	531	0.350	0.353	0.411	0.388	0.439	0.653	0.500
TSLA	124	0.563	0.494	0.573	0.589	0.556	0.556	0.502
FB	26	0.500	0.469	0.462	0.538	0.577	0.577	0.495

Volatility task, 'paper' labels

Stock	n test	NN (report)	NN (tuned)	LogReg (report)	LogReg (tuned)	Ensemble (new)	Majority	Random
GME	1462	0.256	0.275	0.272	0.271	0.503	0.805	0.500
AMC	531	0.869	0.960	0.902	0.902	0.960	0.960	0.499
TSLA	124	0.474	0.563	0.427	0.460	0.855	0.855	0.500
FB	26	0.508	0.546	0.692	0.692	0.500	0.500	0.507

Leakage check (report's direction task, report configuration): with a *shuffled* split the network scores GME 0.77, AMC 0.78, TSLA 0.53, FB 0.68; on our *chronological* split it scores GME 0.57, AMC 0.35, TSLA 0.60, FB 0.47. Same-day posts share a label, so shuffling leaks the answer.

New model over all 16 settings (both label modes, both tasks): the shrinkage ensemble matches or beats the majority baseline in **11/16** (best tuned old model: 6/16) and matches or beats both tuned old models in **12/16**. Mean accuracy: ensemble 0.605, tuned LogReg 0.542, tuned NN 0.520, majority 0.685. Mean test log-loss: ensemble 0.639, LogReg 0.667, NN 0.822, majority 0.662. CV set $\lambda = 0$ (pure prior) in 4 settings.



6. Conclusions

(i) **Honest evaluation matters.** We could not reproduce the report's high accuracies (GME 0.84, AMC 0.82) on a time-ordered split; a shuffled split inflates AMC and FB, which suggests leakage is a likely cause (we cannot verify the report's exact split). On a future hold-out no direction model beats the majority-class baseline by more than the test-set standard error (see table; AMC is far below it), so we cannot claim WSB post features predict next-day direction from these 7 features. (ii) **Tuning helps only modestly:** for AMC volatility the tuned network improves from 0.87 to 0.96 (only matching the trivial majority baseline 0.96), and for TSLA from 0.47 to 0.56 (still far below its majority baseline 0.85); for the rest it mostly selects the trivial 'predict the majority' solution. (iii) **The new shrinkage ensemble is the most reliable model**: it reaches the majority baseline in 11/16 settings and lifts the hardest cases most (GME volatility 0.50 vs 0.28, AMC direction 0.44 vs 0.39, TSLA volatility 0.85 vs 0.56). Its strength is knowing when the features do *not* generalise (it then falls back to the prior); its AUC stays near 0.5, so it still finds no real predictive signal in the posts. (iv) The back-test confirms it: the network strategy stays flat on GME/AMC/TSLA and slightly under-performs always-long on FB. (v) Distribution shift is the main obstacle: post volume peaks during the Jan-Feb 2021 squeeze, so train and test regimes differ; TSLA and FB test sets are tiny (124 and 26 posts, std. error 5-10 pp), so their differences are not significant. **Future work:** richer text features (BERT embeddings), day-level aggregation, more data, and a transaction-cost-aware strategy.